CUSTOMER LIFETIME VALUE (CLV) ANALYSIS

Project Report

Project focus	Estimate customer value from historical retail transactions
Dataset	Online Retail II — UCI Machine Learning Repository
Tools	Python, Pandas, Matplotlib, Excel, Google Colab
Deliverables	Customer-level table, customer segments, charts, and business recommendations

Note: Add the actual totals, charts, and observations generated by your notebook before submitting the report. No numerical findings are fabricated in this template.

1. Executive Summary

This project uses historical retail transaction data to explore customer purchasing behavior and estimate customer value. The workflow combines two yearly sheets, cleans the records, calculates transaction revenue, summarizes purchases by customer, and groups customers into four revenue-based segments.

The baseline metric annualizes observed revenue over the time between a customer's first and last recorded purchases. It is a simple descriptive estimate, not a validated prediction of future customer lifetime value (CLV).

2. Project Objectives

- Combine the two yearly sheets into one transaction dataset.
- Clean records using documented rules.
- Calculate revenue, distinct order count, average order value, first and last purchase dates, and recency.
- Create a customer-level table with an annualized historical revenue estimate.
- Create four customer segments and summarize each segment.
- Explain business implications, assumptions, and limitations.

3. Dataset Description

The Online Retail II dataset is available from the UCI Machine Learning Repository. The workbook used in this workflow contains the sheets “Year 2009-2010” and “Year 2010-2011”. Common fields include Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country.

Dataset source: https://archive.ics.uci.edu/dataset/502/online+retail+ii

Key fields

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Invoice	Count distinct orders and identify cancellation invoices.
Quantity	Number of units on a transaction line.
InvoiceDate	Transaction date and time.
Price	Unit price used to calculate revenue.
Customer ID	Identifier for customer-level aggregation.
Revenue	Calculated as Quantity × Price.

4. Tools and Technologies

- Python for data preparation and calculations.
- Pandas for loading, cleaning, grouping, and exporting data.

- Matplotlib for charts.
- Excel for sharing and reviewing tables.
- Google Colab for running the notebook.

5. Methodology

5.1 Load and combine data

Read both yearly sheets from the Excel workbook and concatenate them into one transaction-level dataset. Keep the original dataset unchanged so that cleaning decisions can be reviewed.

5.2 Clean the data

- Remove duplicate rows.
- Exclude rows without Customer ID because customer-level metrics cannot be calculated for them.
- Exclude invoice identifiers beginning with “C” as cancellation records.
- Keep rows with positive quantities and positive prices for this revenue analysis.
- Convert InvoiceDate to datetime and exclude rows with invalid dates.

These rules define the analysis population. Different choices can change the results and should be documented.

5.3 Calculate transaction revenue

For each transaction line, Revenue = Quantity × Price. Customer total revenue is the sum of eligible line revenue for that customer. Revenue is not the same as profit.

5.4 Calculate customer metrics

- Total Revenue: sum of eligible transaction revenue per customer.
- Total Orders: number of distinct invoice identifiers per customer.
- Average Order Value: Total Revenue ÷ Total Orders.
- First Purchase and Last Purchase: earliest and latest observed transaction dates.
- Recency: days between the last observed purchase and the selected analysis date.

5.5 Estimate annualized historical revenue

Estimated Annualized Revenue = Total Revenue ÷ Observed Purchase Period in Days × 365. The observed purchase period is the difference between the customer's first and last purchase dates; a minimum denominator of one day is used to avoid division by zero.

This measure annualizes historical revenue over the observed span. It does not explicitly model churn, future repeat-purchase probability, profit margin, discounting, or customer acquisition cost.

5.6 Segment customers

Rank customers by total historical revenue and split them into four approximately equal-sized groups: Low Value, Medium Value, High Value, and Top Value. These are descriptive historical-revenue segments, not predictive classifications.

6. Project Deliverables

- CLV_Table: customer-level revenue, order, timing, recency, and annualized revenue metrics.
- Segment_Analysis: customer count, total revenue, average revenue, and average order count by segment.
- Cleaned_Data: cleaned transaction-level records used for the analysis.
- Python notebook: reproducible workflow and visualizations.

7. Business Insights and Recommendations

- Retention: test loyalty benefits or personalized communication for customers with high historical revenue.
- Repeat purchases: test relevant follow-up offers for customers with low order frequency.

- Revenue concentration: calculate the share of total revenue contributed by the Top Value segment.
- Campaign evaluation: compare outcomes before and after campaigns instead of assuming every offer will work.
- Profitability: if cost or margin data becomes available, estimate customer contribution profit rather than relying only on revenue.

Add the actual numerical results and chart observations from the completed notebook here. Do not claim specific totals or segment trends until they have been calculated.

8. Assumptions and Limitations

- Positive quantities and prices are treated as eligible sales; cancellation invoices beginning with “C” are excluded.
- Revenue is used as a proxy for customer value, but it is not profit.
- Customers with one purchase or a very short observed purchase span may have unusually high annualized estimates.
- The dataset observation window limits conclusions about long-term behavior.
- The segments are based on historical revenue rather than a validated predictive CLV model.
- Acquisition costs, margins, discount rates, and future retention probabilities are not included.
- Returns and other dataset-specific business rules may need further review for business use.

9. Future Improvements

- Compare this baseline with a repeat-purchase or probabilistic CLV method.
- Include gross margin and acquisition cost to estimate customer profitability.
- Use a calibration period and a holdout period to test predictions.
- Add RFM analysis and cohort retention analysis.
- Automate data-quality checks and reporting charts.

10. Conclusion

This project demonstrates a practical workflow for transforming retail transaction data into customer-level purchasing metrics and revenue-based segments. The results can support exploratory customer analysis and help prioritize retention experiments. The annualized historical revenue measure should be treated as a baseline indicator; a reliable predictive CLV model requires explicit assumptions about future purchases, retention, margins, and costs.

11. References

UCI Machine Learning Repository, Online Retail II dataset: https://archive.ics.uci.edu/dataset/502/online+retail+ii

Pandas documentation: https://pandas.pydata.org/docs/

Matplotlib documentation: https://matplotlib.org/stable/